

IE 203 - Operations Research II
Quiz II

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Duration: 45 minutes
This is a CLOSED BOOK exam.

Question I (60 pts.)

XYZ is considering four projects. Project 1 will yield a profit of 16 millions TL; project 2, a profit of 22 millions TL; project 3, a profit of 12 millions TL; and project 4, profit of 8 millions TL. Each project requires a certain investment: project 1, 5 millions TL; project 2, 7 millions TL; project 3, 4 millions TL; project 4, 3 millions TL. Currently, the available budget is 14 millions TL.

- a. Formulate a knapsack problem whose solution will provide XYZ an investment plan maximizing the total profit obtained from projects 1 - 4.
- b. Solve the formulation and find an optimal solution using branch-and-bound with last-in-first-out rule. Show all your work and prepare the branch-and-bound search tree.

Question II (40 pts.)

Find the stationary points of the following problem using **KKT first order necessary conditions**.

$$\begin{array}{ll}\min & (2x_1 - 1)^2 + x_2^2 \\ \text{s.t.} & -2x_1^2 + x_2^2 = 0 \\ & x_1 \geq -1, x_2 \geq -2\end{array}$$